BIMM-143: INTRODUCTION TO BIOINFORMATICS

The find-a-gene project assignment
http://thegrantlab.org/bimm143 Dr.
Barry Grant

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Overview:

The find-a-gene project is a required assignment for BIMM-143. You should prepare a written report in **PDF** format that has responses to each question labeled **[Q1] - [Q10]** below. You may wish to consult the scoring rubric at the end of this document and the example report provided online (note that the example report is from a *previous quarter* and the questions may differ).

The objective with this assignment is for you to demonstrate your grasp of database searching, sequence analysis, structure analysis and the R environment that we have covered in class.

Due Date:

Your responses to questions Q1-Q4 are due at 12pm on the **Monday of Week 5** (see the Assignments and Grading section of our website for details). Note that these first set of answers can be obtained very quickly (at best within 15 or 20 minutes), so if you don't succeed at first, just keep trying.

The complete assignment, including responses to all questions, is due at 12pm on the **Monday of Week 10**.

Submission instructions:

Your report formatted as a **PDF document** should be uploaded to ***GradeScope***. Please make sure to include your UCSD email and PID number on the first page.

Be sure to include your UCSD email and PID number on the first page of your report.

Submit your preliminary report with answers to Q1-Q4 as soon as you can so we can determine if you have found a novel gene. Submit this preliminary report as one document with screen shots of the results inserted appropriately.

See the demonstration report linked to on the course website for an example of format. Note again that example questions may differ. I will indicate on GradeScope my decision (1pt indicating all is good, 0pts revisions required). You should proceed with subsequent questions only after we are sure you have found a novel gene (and thus be successful in the later stages of the project).

For the final report add your results for Q5-Q10 to the preliminary report and submit the final document containing your results for all questions.

Please do not send only Q5-Q10 answers as the final report.

j5aquino@ucsd.edu
PID: A17756721

Questions:

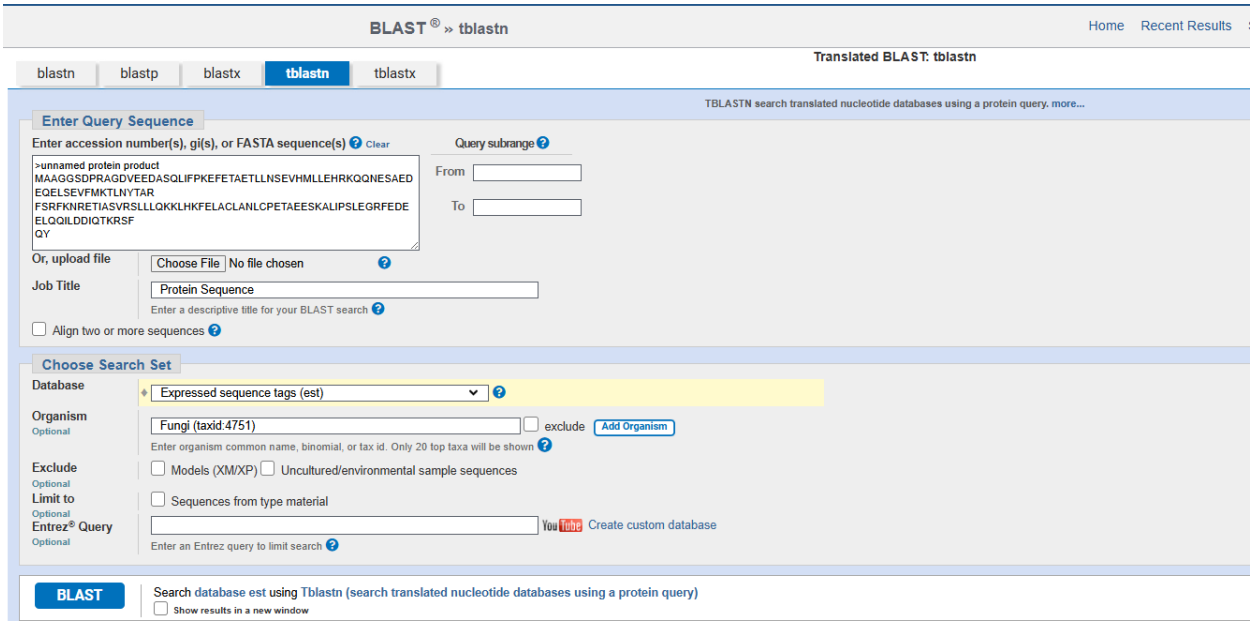
[Q1] Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as it's function is known.
If you do not have a favorite protein, select human RBP4 or KIF11. Do not use beta globin as this is in the worked example report that I provide you with online.

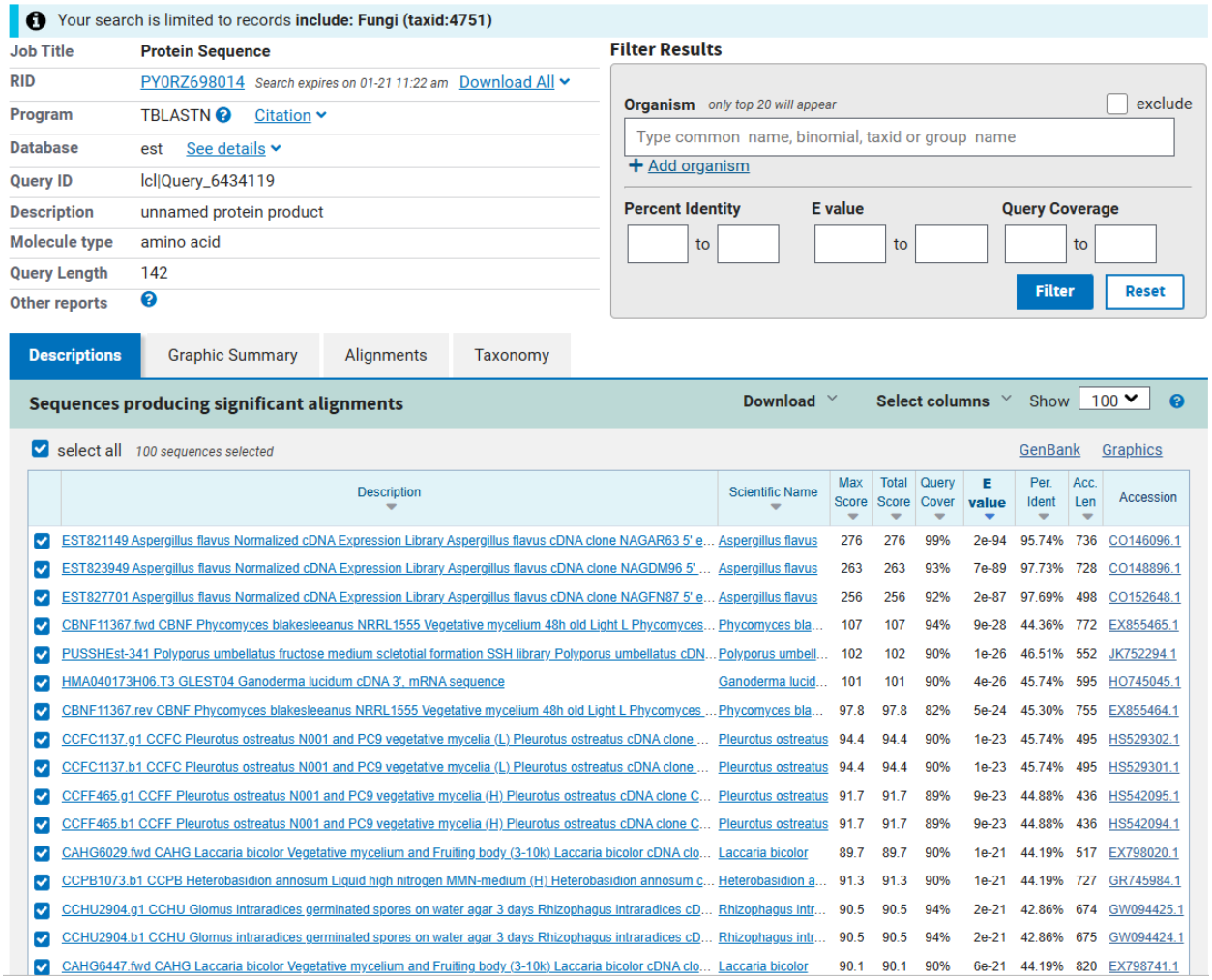
Name: Polymerase (RNA) II (DNA directed) polypeptide D [Homo sapiens]
Species: Homo Sapiens
Accession Number: AAH93797
Known Function: Involved in transcription by RNA polymerase II, DNA-templated transcription initiation, and transcription initiation at RNA polymerase II promoter

[Q2] Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Method: tblastn
Database: Expressed sequence tags (est)
Limits: Organism: Fungi (taxid: 4751)

Also include the output of that BLAST search in your document. If appropriate, change the font to `Courier size 10` so that the results are displayed neatly. You can also screen capture a BLAST output (e.g. alt print screen on a PC or on a MAC press ⌘-shift-4. The pointer becomes a bulls eye. Select the area you wish to capture and release. The image is saved as a file called `Screen Shot [].png` in your Desktop directory). It is **not** necessary to print out all of the blast results if there are many pages.

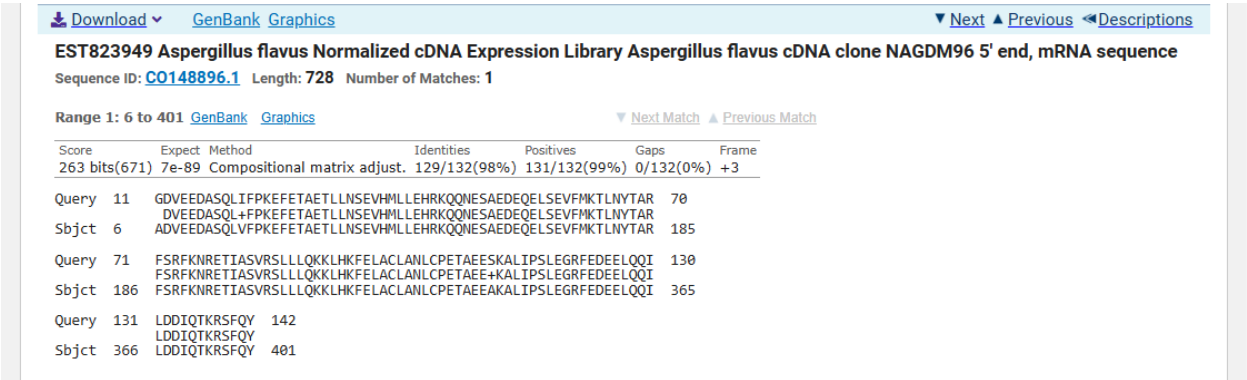




On the BLAST results, clearly indicate a match that represents a protein sequence, encoded from some DNA sequence, that is homologous to your query protein. I need to be able to inspect the pairwise alignment you have selected, including the E value and score. It should be labeled a "genomic clone" or "mRNA sequence", etc. - but include no functional annotation.

Chosen match:

Accession Number: CO148896.1
Species: Aspergillus flavus



Alignment details:

>CO148896.1 EST823949 Aspergillus flavus Normalized cDNA Expression Library Aspergillus flavus cDNA clone NAGDM96 5' end, mRNA sequence

Length = 728

Score = 263 bits (671), Expect = 7e-89, Method: Compositional matrix adjust.

Identities = 129/132 (99%), Gaps = 0/132 (0%), Frame = +3

```
Query  11   GDVEEDASQLIFPKEFETAETLLNSEVHMLLEHRKQQNESAEDEQELSEVFMKTLNYTAR  70
            DVEEDASQL+FPKEFETAETLLNSEVHMLLEHRKQQNESAEDEQELSEVFMKTLNYTAR
Sbjct  6    ADVEEDASQLVFPKEFETAETLLNSEVHMLLEHRKQQNESAEDEQELSEVFMKTLNYTAR  185

Query  71   FSRFKNRETIASVRSLLLQKKLHKFELACLANLCPETAEESKALIPSLEGRFEDEELQQI  130
            FSRFKNRETIASVRSLLLQKKLHKFELACLANLCPETAEE+KALIPSLEGRFEDEELQQI
Sbjct  186  FSRFKNRETIASVRSLLLQKKLHKFELACLANLCPETAEEAKALIPSLEGRFEDEELQQI  365

Query  131  LDDIQTKRSFQY  142
            LDDIQTKRSFQY
Sbjct  366  LDDIQTKRSFQY  401
```

In general, [Q2] is the most difficult for students because it requires you to have a “feel” for how to interpret BLAST results. You need to distinguish between a perfect match to your query (i.e. a sequence that is not “novel”), a near match (something that might be “novel”, depending on the results of [Q4]), and a non-homologous result.

If you are having trouble finding a novel gene try restricting your search to an organism that is poorly annotated.

[Q3] Gather information about this “novel” **protein**. At a minimum, show me the protein sequence of the “novel” protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don’t forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don’t have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format.

Sequence (FASTA alignment sequence translated using EMBOSS Transeq):

```
>6-401_1 EST823949 Aspergillus flavus Normalized cDNA Expression
Library Aspergillus flavus cDNA clone NAGDM96 5' end, mRNA
sequence
```

```
ADVEEDASQLVFPKEFETAETLLNSEVHMLLEHRKQQNESAEDEQELSEVFMKTLNYTAR
FSRFKNRETIASVRSLLLQKKLHKFELACLANLCPETAEEAKALIPSLEGRFEDEELQQI
LDDIQTKRSFQY
```

Here, tell me the name of the novel protein, and the species from which it derives. It is very unlikely (but still definitely possible) that you will find a novel gene from an organism such as *S. cerevisiae*, human or mouse, because those genomes have already been thoroughly annotated. It is more likely that you will discover a new gene in a genome that is currently being sequenced, such as bacteria or plants or protozoa.

Name: EST823949 Aspergillus flavus Normalized cDNA Expression Library
Aspergillus flavus cDNA clone NAGDM96 5' end, mRNA sequence
Species: Aspergillus flavus

[Q4] Prove that this gene, and its corresponding protein, are novel. For the purposes of this project, “novel” is defined as follows. Take the protein sequence (your answer to [Q3]), and use it as a query in a blastp search of the nr database at NCBI.

- If there is a match with 100% amino acid identity to a protein in the database, from the same species, then your protein is NOT novel (even if the match is to a protein with a name such as “unknown”). Someone has already found and annotated this sequence, and assigned it an accession number.
- If the top match reported has less than 100% identity, then it is likely that your protein is novel, and you have succeeded.
- If there is a match with 100% identity, but to a different species than the one you started with, then you have likely succeeded in finding a novel gene.
- If there are no database matches to the original query from [Q1], this indicates that you have partially succeeded: yes, you may have found a new gene, but no, it is not actually homologous to the original query. You should probably start over.

The blastp results show numerous species with a 100% Percent Identity to the gene that are not Aspergillus flavus, thus the gene and its corresponding protein are highly likely to be novel.
The following screenshots show a portion of the blastp search results, the alignment of the top two search results, and the blastp search parameters used.

DNA-directed RNA polymerase II subunit RPB4 [Emydura macquarii macquarii]	Emydura macquarii macquarii	266	266	100%	3e-89	100.00%	142	XP_067394953.1
DNA-directed RNA polymerase II subunit RPB4 [Agelaius tricolor]	Agelaius tricolor	270	270	100%	3e-89	100.00%	240	XP_071293280.1
DNA-directed RNA polymerase II subunit RPB4 [Chelonoidis abingdonii]	Chelonoidis abingdonii	266	266	100%	3e-89	100.00%	142	XP_032620687.1
PREDICTED: DNA-directed RNA polymerase II subunit RPB4 [Gavialis gangeticus]	Gavialis gangeticus	266	266	100%	3e-89	100.00%	142	XP_019359266.1
DNA-directed RNA polymerase II subunit RPB4 [Lepidochelys kempii]	Lepidochelys kempii	266	266	100%	4e-89	100.00%	142	XP_073216257.1
RPB4 polymerase [Nyctibius bracteatus]	Nyctibius bracteatus	266	266	100%	4e-89	100.00%	143	NXF36936.1
DNA-directed RNA polymerase II subunit RPB4 [Alligator mississippiensis]	Alligator mississippiensis	266	266	100%	4e-89	100.00%	142	XP_059586883.1
DNA-directed RNA polymerase II subunit RPB4 [Apus apus]	Apus apus	266	266	100%	4e-89	100.00%	142	XP_051482859.1
DNA-directed RNA polymerase II subunit RPB4 [Pelodiscus sinensis]	Pelodiscus sinensis	266	266	100%	4e-89	100.00%	142	XP_075793570.1
DNA-directed RNA polymerase II subunit RPB4 [Gopherus evgoodei]	Gopherus evgoodei	266	266	100%	4e-89	100.00%	142	XP_030431933.1
DNA-directed RNA polymerase II subunit RPB4 [Terrapene triunguis]	Terrapene triunguis	266	266	100%	4e-89	100.00%	142	XP_024072598.1
RPB4 polymerase [Grallaria varia]	Grallaria varia	266	266	100%	4e-89	100.00%	143	NXG24128.1
DNA-directed RNA polymerase II subunit RPB4 [Caretta caretta]	Caretta caretta	266	266	100%	4e-89	100.00%	142	XP_048721080.1
DNA-directed RNA polymerase II subunit RPB4 [Gymnogyps californianus]	Gymnogyps californianus	266	266	100%	5e-89	100.00%	142	XP_050758592.1
RPB4 polymerase [Catharus fuscescens]	Catharus fuscescens	266	266	100%	5e-89	100.00%	143	NXQ44703.1
RPB4 polymerase [Chunga burmeisteri]	Chunga burmeisteri	266	266	100%	5e-89	100.00%	143	NWS51081.1
RPB4 polymerase [Ardeotis kori]	Ardeotis kori	266	266	100%	5e-89	100.00%	143	NXE27591.1
RPB4 polymerase [Mionectes macconnelli]	Mionectes macconnelli	265	265	100%	5e-89	100.00%	143	NWT00978.1

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EST821149 Aspergillus flavus Normalized cDNA Expression Library Aspergillus flavus cDNA clone NAGAR63 5' end similar to DNA-directed RNA polymerase II 16 kDa polypeptide (EC 2.7.7.6) (RPB4). [Human] {Homo sapiens}, mRNA sequence

Sequence ID: CO146096.1 Length: 736 Number of Matches: 1

Range 1: 5 to 427 GenBank Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
276 bits(707)	2e-94	Compositional matrix adjust.	135/141(96%)	138/141(97%)	0/141(0%)	+2

```
Query  2    AAGGSDPRAGDVEEDASQLIFPKEFETAETLLNSEVHMLLEHRKQQNESAEDEQELSEVF  61
            +AGGSDPR  DVEEDASQL+FPKEFETAETLLNSEVHMLLEHRKQQNESAEDEQELSEVF
Sbjct  5    SAGGSDPRTADVEEDASQLVFPKEFETAETLLNSEVHMLLEHRKQQNESAEDEQELSEVF  184

Query  62   MKTLNYTARFSRFKNRETIASVRSLLLQKKLHKFELACLANLCPETAEESKALIPSLEGR  121
            MKTLNYTARFSRFKNRETIASVRSLLLQKKLHKFELACLANLCPETAEE+KALIPSLEGR
Sbjct  185  MKTLNYTARFSRFKNRETIASVRSLLLQKKLHKFELACLANLCPETAEEAKALIPSLEGR  364

Query  122  FEDEELQQILDDIQTKRSFQY  142
            FEDE LQQILDDIQTKRSFQY
Sbjct  365  FEDE*LQQILDDIQTKRSFQY  427
```

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EST823949 Aspergillus flavus Normalized cDNA Expression Library Aspergillus flavus cDNA clone NAGDM96 5' end, mRNA sequence

Sequence ID: CO148896.1 Length: 728 Number of Matches: 1

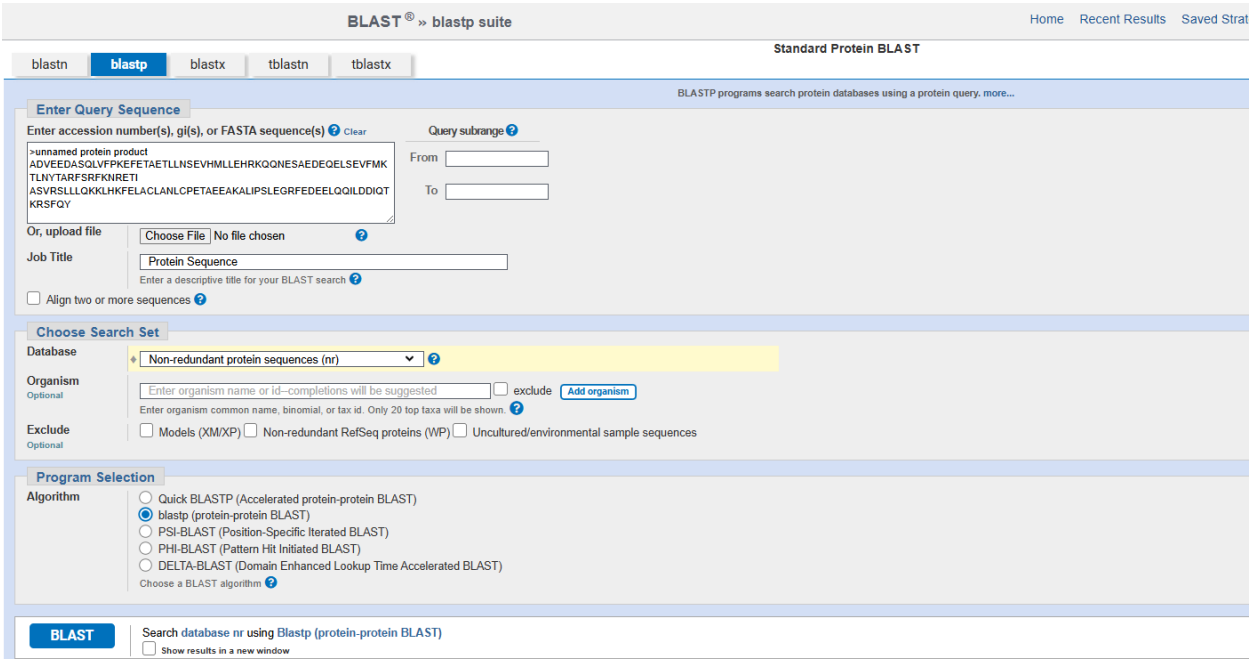
Range 1: 6 to 401 GenBank Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
263 bits(671)	7e-89	Compositional matrix adjust.	129/132(98%)	131/132(99%)	0/132(0%)	+3

```
Query  11   GDVEEDASQLIFPKEFETAETLLNSEVHMLLEHRKQQNESAEDEQELSEVFMKTLNYTAR  70
             DVEEDASQL+FPKEFETAETLLNSEVHMLLEHRKQQNESAEDEQELSEVFMKTLNYTAR
Sbjct  6    ADVEEDASQLVFPKEFETAETLLNSEVHMLLEHRKQQNESAEDEQELSEVFMKTLNYTAR  185

Query  71   FSRFKNRETIASVRSLLLQKKLHKFELACLANLCPETAEESKALIPSLEGRFEDEELQQI  130
            FSRFKNRETIASVRSLLLQKKLHKFELACLANLCPETAEE+KALIPSLEGRFEDEELQQI
Sbjct  186  FSRFKNRETIASVRSLLLQKKLHKFELACLANLCPETAEEAKALIPSLEGRFEDEELQQI  365

Query  131  LDDIQTKRSFQY  142
            LDDIQTKRSFQY
Sbjct  366  LDDIQTKRSFQY  401
```



[Q5] Generate a multiple sequence alignment with your novel protein, your original query protein, and a group of other members of this family from different species. A typical number of proteins to use in a multiple sequence alignment for this assignment purpose is a minimum of 5 and a maximum of 20 - although the exact number is up to you. Include the multiple sequence alignment in your report. Use Courier font with a size appropriate to fit page width.

Side-note: Indicate your sequence in the alignment by choosing an appropriate name for each sequence in the input unaligned sequence file (i.e. edit the sequence file so that the species, or short common, names (rather than accession numbers) display in the output alignment and in the subsequent answers below). The goal in this step is to create an interesting an alignment for building a phylogenetic tree that illustrates species divergence.

Re-labeled Sequences used for Alignment (Common Names for Species used):

```
> Humans|AAH93797.1 Polymerase (RNA) II (DNA directed) polypeptide D [Homo
sapiens]
MAAGGSDPRAGDVEEDASQLIFPKEFETAETLLNSEVHMLLEHRKQQNESAEDEQELSEVFMKTLNYTARFSRFKNRE
TIASVRSLLLQKKLHKFELACLANLCPETAEESKALIPSLEGRFEDEELQQILDDIQTKRSFQY


> Yellow Mold|CO148896.1_3 EST823949 Aspergillus flavus Normalized cDNA
Expression Library Aspergillus flavus cDNA clone NAGDM96 5' end, mRNA
sequence
TADVEEDASQLVFPKEFETAETLLNSEVHMLLEHRKQQNESAEDEQELSEVFMKTLNYTARFSRFKNRETIASVRSLL
LQKKLHKFELACLANLCPETAEEAKALIPSLEGRFEDEELQQILDDIQTKRSFQY*S*PSTPPNEQKI*GLCFSTLC
TNPAPGLSSG*RELQCHTLQAYWRSVKSSCFDKDTCIVRALELEKTFSWWLKYLFLPHSSPQFGSFIHVSVLWQKCR
PVWVVGFKSG
```

> Umbrella Polypore|JK752294.1_2 PUSSHEst-341 Polyporus umbellatus fructose medium scletotial formation SSH library Polyporus umbellatus cDNA similar to DNA-directed RNA polymerase II, putative, mRNA sequence
HGGNPLVLATMALRHRPAYEEEDASALRLGSEFNNAGCLLISEVRYLLEHRPSEKAAPDTTVYNKTLDYVKTFAKFDK
PEAAAAIRETLRRESALTQFEIAQIANLCPVTAEEAKSIIPSLVKIEDDKLQPLLDEIQTMRKFQS*RQIIASLFDG
ISQCGSSAQCFVAMSSQHHGLVGYRASSV

> Oyster Mushroom|HS542094.1_1 CCFF465.b1 CCFF Pleurotus ostreatus N001 and PC9 vegetative mycelia (H) Pleurotus ostreatus cDNA clone CCFF465 5', mRNA sequence
DASALKLGPEFNNAGCLLISEVKYLLENRDKDAPDTPVYNKTLDYVKTFTKFSTTDSASAVRETLRREPALTQFETAQ
IANLCPADAEEAKSIIPSLMKIEDDRLQALLDEIQTMRKFQT*STHYCQFVV*PALHIYTPIQLIHWX

> AMF (Arbuscular Mucorrhizal Fungus)|GW094424.1_3 CCHU2904.b1 CCHU Glomus intraradices germinated spores on water agar 3 days Rhizophagus intraradices cDNA clone CCHU2904 5', mRNA sequence
LFSMSHHAPIIHRSRKAPQEEEDAAKLKFGEDFEDEEFLFISEVALLLEKIPDSQKNNTVYKKTEELVNTFTRFHNDE
SVRELRKTLMRHKLHKYELAQLANLVCEEIDEAKSLIPSLAKRSDEEIRAMLDDISALKKYQS*KMT*ESKSS**IR
YFEVFYFLHIFNI*SSILRVLRQDNN*KSWHLPVIVIIYP*RRCLKPRNFFIRNKNFL*RDTYRIT*LFX

> Fission Yeast|FY082517.1_1 SPA (pGCAP10) cDNA library Schizosaccharomyces pombe cDNA clone spa59b23 5', mRNA sequence
IETRSNSKKAGCCA*RAF*HYFWEYICIIVIDLIR*KEKTTF*RECLRASIHNAEAIFEEDAAQLKLGPEFENEDMLT
VSEAKILIETVLAQRARETNGEIPMTDVMKKTVAYFNVFARFKTAEATYACERILGNRFHKFERAQLGTLCCEDAEE
ARTLIPSLANKIDDQNLQGILDELSTLRKFQD*

> Soil Borne Fungus|HG362892.1_2 Rhizoctonia solani EST, clone rsol_all_isotig03328, strain AG1-IB, mRNA sequence
*FPSRATNDTIT*TRDQEMATRSRRTQNEEEDAAALKLGSEFNNAGCLLISEVKYLLEHREKDSPDTAVYNKALEYVK
TFTKFNTNESSSAVREVLRREPNLTQFETAQIANLCPADVEEARNIIPSLVKIDEDRLQTLLDEVQALRRFQT*LCC
PYAWLSVVAFHISSYIHLCHLSSGSYPPALSLPQSTRPTQL

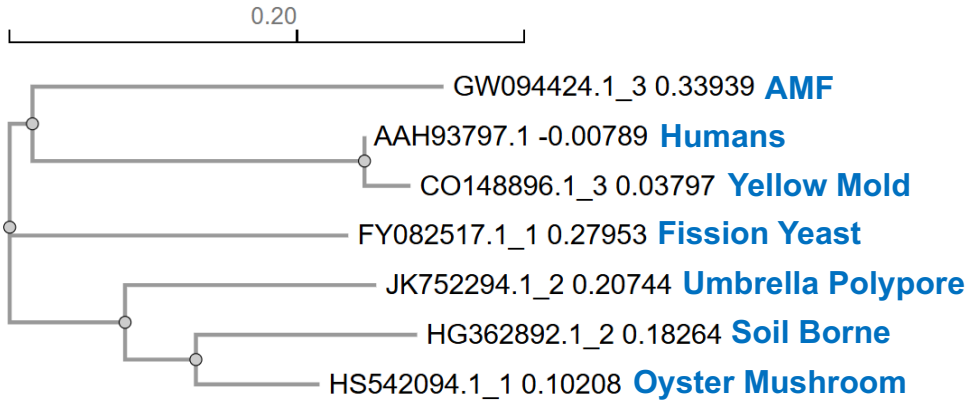
Multiple Sequence Alignment using MUSCLE (3.8)

```
AMF                ---EEDAAKLKFGEDFEDEEFLFISEVALLLEK--------IPDSQKNNTVYKKTEEL
Humans             ---EEDASQLIFPKEFETAETLLNSEVHMLLEH-RKQQNESAEDEQELSEVFMKTLNY
Yellow Mold        ---EEDASQLVFPKEFETAETLLNSEVHMLLEH-RKQQNESAEDEQELSEVFMKTLNY
Fission Yeast      AIFEEDAAQLKLGPEFENEDMLTVSEAKILIETVLAQRARETNGEIPMTDVMKKTVAY
Umbrella Polypore  ---EEDASALRLGSEFNNAGCLLISEVRYLLEH------RPSEKAAPDTTVYNKTLDY
Soil Borne         ---EEDAAALKLGSEFNNAGCLLISEVKYLLEH--------REKDSPDTAVYNKALEY
Oyster Mushroom    -----DASALKLGPEFNNAGCLLISEVKYLLEN--------RDKDAPDTPVYNKTLDY

AMF                VNTFTRFHNDESVRELRKTLMRH-KLHKYELAQLANLVCEEIDEAKSLIPSLA-KRSD
Humans             TARFSRFKNRETIASVRSLLLQK-KLHKFELACLANLCPETAEESKALIPSLEGRFED
Yellow Mold        TARFSRFKNRETIASVRSLLLQK-KLHKFELACLANLCPETAEEAKALIPSLEGRFED
Fission Yeast      FNVFARFKTAEATYACERILGN--RFHKFERAQLGTLCCEDAEEARTLIPSLANKIDD
Umbrella Polypore  VKTFAKFDKPEAAAAIRETLRRESALTQFEIAQIANLCPVTAEEAKSIIPSLV-KIED
Soil Borne         VKTFTKFNTNESSSAVREVLRREPNLTQFETAQIANLCPADVEEARNIIPSLV-KIDE
Oyster Mushroom    VKTFTKFSTTDSASAVRETLRREPALTQFETAQIANLCPADAEEAKSIIPSLM-KIED

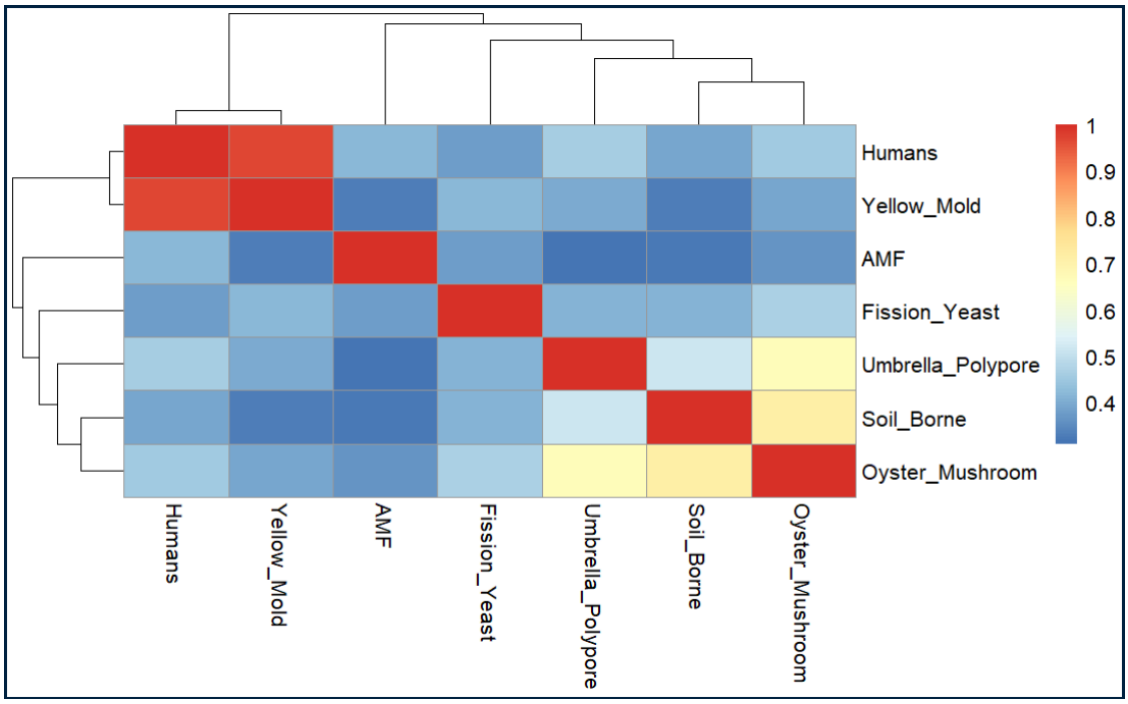
AMF                EEIRAMLDDISALKKYQSKMTESKSSIRYFEVFYFLHIFNI----------------
Humans             EELQQILDDIQTKRSFQY-------------------------------------
Yellow Mold        EELQQILDDIQTKRSFQYSPSTPPNEQKIGLCFSTLCTNPAPGLSSGRELQCHTLQAY
Fission Yeast      QNLQGILDELSTLRKFQD-------------------------------------
Umbrella Polypore  DKLQPLLDEIQTMRKFQS-----RQIIASLFDGISQCGSSA----------------
Soil Borne         DRLQTLLDEVQALRRFQT--------LCCPYAWLSVVAFHI----------------
Oyster Mushroom    DRLQALLDEIQTMRKFQT---------STHYCQFVVPALHI----------------
```

[Q6] Create a phylogenetic tree, using either a parsimony or distance-based approach. Bootstrapping and tree rooting are optional. Use “simple phylogeny” online from the EBI or any respected phylogeny program (such as MEGA, PAUP, or Phylip). Paste an image of your Cladogram or tree output in your report.



[Q7] Generate a sequence identity based **heatmap** of your aligned sequences using R.

If necessary convert your sequence alignment to the ubiquitous FASTA format (Seaview can read in clustal format and “Save as” FASTA format for example). Read this FASTA format alignment into R with the help of functions in the **Bio3D package**. Calculate a sequence identity matrix (again using a function within the Bio3D package). Then generate a heatmap plot and add to your report. Do make sure your labels are visible and not cut at the figure margins.



[Q8] Using R/Bio3D (or an online blast server if you prefer), search the main protein structure database for the most similar atomic resolution structures to your aligned sequences.

List the top 3 *unique* hits (i.e. not hits representing different chains from the same structure) along with their Evalue and sequence identity to your query. Please also add annotation details of these structures. For example include the annotation terms PDB identifier (structureId), Method used to solve the structure (experimentalTechnique), resolution (resolution), and source organism (source).

HINT: You can use a single sequence from your alignment or generate a consensus sequence from your alignment using the Bio3D function consensus(). The Bio3D functions blast.pdb(), plot.blast() and pdb.annotate() are likely to be of most relevance for completing this task. Note that the results of blast.pdb() contain the hits PDB identifier (or pdb.id) as well as Evalue and identity. The results of pdb.annotate() contain the other annotation terms noted above.

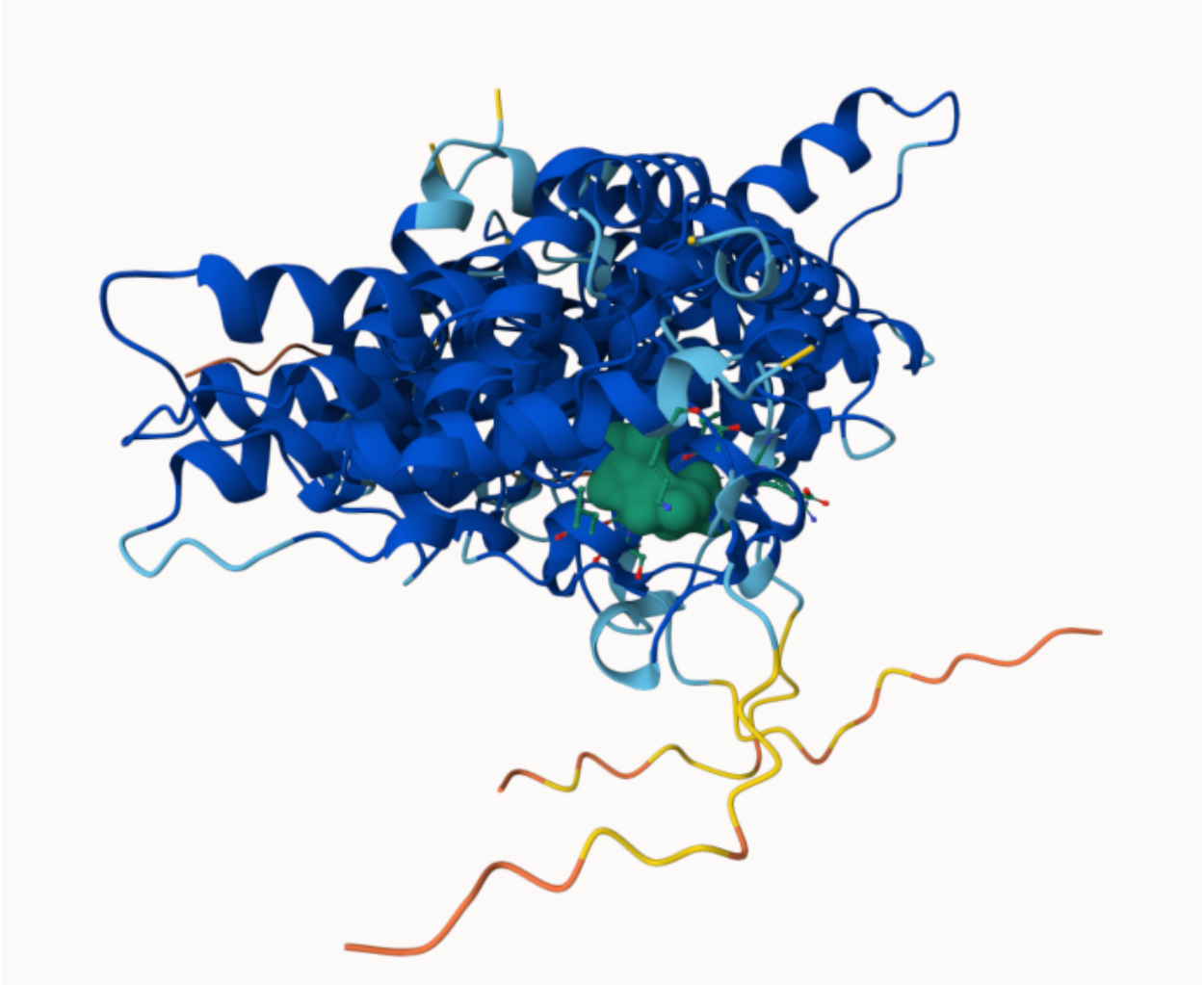
Note that if your consensus sequence has lots of gap positions then it will be better to use an original sequence from the alignment for your search of the PDB. In this case you could chose the sequence with the highest identity to all others in your alignment by calculating the row-wise maximum from your sequence identity matrix.

ID	Technique	Resolution	Source	E-value	Identity
2C35	X-Ray Diffraction	2.70	Homo sapiens	3e-98	100%
5FLM	Electron Microscopy	3.40	Bos taurus	2e-97	99.30%
7OKX	Electron Microscopy	3.30	Sus scrofa	2e-97	100%

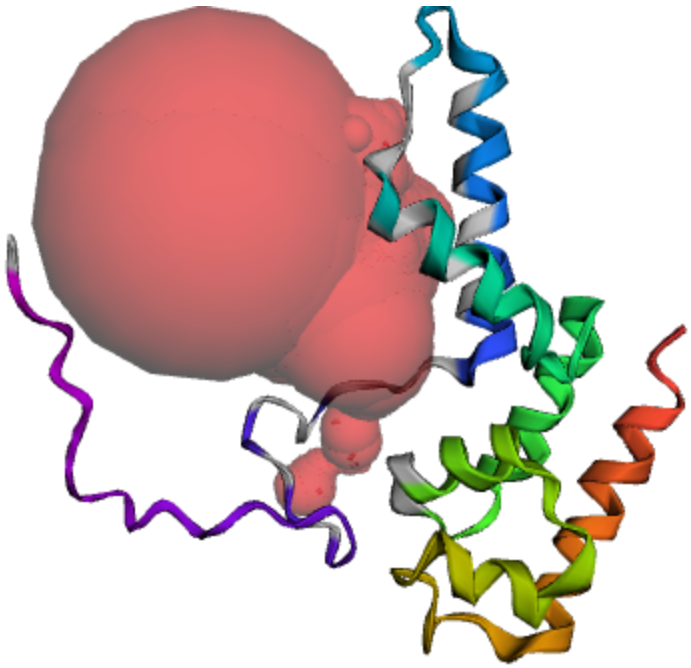
[Q9] Using AlphaFold notebook generate a structural model using the default parameters for your novel protein sequence.

> Note that this can take some time depending upon your sequence length. If your model is taking many hours to generate or your input sequence yields a “too many amino acids” (i.e. length) error you can focus on a single domain from your sequence - identify region by searching for PFAM domain matches.

Once complete save the resulting PDB format file for your records. Finally, generate a molecular figure of your generated PDB structure using the **Mol* viewer** online (or VMD/PyMol/Chimera if you prefer). To complete your analysis you should highlight *conserved residues* that are likely to be functional as **spacefill** and the protein as **cartoon** colored by local alpha fold *pLDDT quality score*. You can determine conserved residues from the alignment generated by the AlphaFold server and use a conservation cutoff appropriate for the diversity of your protein alignment (e.g. between 60% and 99% conserved). Note that *pLDDT* score is contained in the B-factor column of your PDB downloaded file. Please use a white or transparent background for your figure (i.e. not the default black in PyMol/VMD/Chimera etc.).



[Q10] (i) Using your computed structure model (or your closest homologue of known structure from the PDB) predict and locate potential small molecule binding sites using the CASTpFold server (https://cfold.bme.uic.edu/castpfold/). Provide an image or screen-shot of your largest predicted pockets “negative volume” and provide it’s **area** and **volume**.



The largest predicted pocket has an area of 427.729 A^2 and a volume of 2131.938 A^3.

(ii) Perform a “Target” search of ChEMBEL (https://www.ebi.ac.uk/chembl/) with your novel sequence. Are there any **Target Associated Assays** and **ligand efficiency data** reported that may be useful starting points for exploring potential inhibition of your novel protein? If there are no assays listed here simply list “non available as of [date]”.

The novel sequence originating from humans has non available as of 03-02-2026. However, the novel sequence originating from yellow mold has one functional assay (CHEMBL5249696) but no ligand efficiency data.
https://www.ebi.ac.uk/chembl/search_results/CO148896.1

(iii) Briefly discuss (100 words max) the **druggability** of your novel protein based on:

- Presence of well-defined pockets (output of tools like CASTpFold),
- Existence of known inhibitors for related proteins (your search of ChEMBEL),
- Conservation of binding sites across homologs (your conservation analysis in Q10),
- Potential therapeutic applications if this protein were targeted (you can use ChatGPT, Claude etc. backed up by your reading of the literature here).

The novel protein likely has a high druggability. CASTpFold shows its largest pocket volume to be 2131.98 A^3, allowing the drug to easily bind. The assay from ChEMBEL shows that induction may be involved, potentially allowing for transcription of the drug if it's an inducer. Q10 shows that most of the protein is conserved with >90% plDDT, making the binding sites more conserved and potentially easier to target. If the protein was targeted, it could be used towards better controlling the rate of transcription in cell growth via RNA polymerase 2, influencing the potential for cell disease and cancer.

Scoring Rubric: [60 total points available]

Q1 (4 points)	
Protein name	1
Species	1
Accession number	1
Function known	1
Q2 (6 points)	
Blast method	1
Database searched	1
Limits applied	1
Search output list (top hits)	1
Alignment of choice	1
Evalue and other alignment stats	1
Q3 (3 points)	
Protein sequence of choice matches Subject above	1
Name in header	1
Species	1
Q4 (3 point)	
Blastp output list with identities & Evalue	1
Top alignment shown with alignment statistics	1
Results indicates a “novel” gene found	1
Q5 (3 points)	
MSA labeled with useful names	1
MSA trimmed appropriately (i.e. no gap overhangs)	1
Pasted MSA fits report page width (i.e. font, format)	1
Q6 (1 point)	
Figure illustrates sequence clustering pattern	1
Q7 (10 points)	

Heatmap figure included in report	5
Heatmap is legible (i.e. no labels obscured)	5
Q8 (10 points)	
PDB identifiers from multiple species reported	5
Annotation of PDB source, resolution and technique	4
Annotation of Evalue and Sequence Identity	1
Q9 (10 points)	
Structure figure provided	2
Uses white background for molecular figure	1
Figure of high resolution (i.e. not just snapshot)	1
Conserved residues as spacefill	3
Protein cartoon colored by pLDDT quality score	3
Q10 (10 points)	
i) Binding site image, volume and area.	3
ii) Evidence of ChEMBEL searches	1
iii) Druggability discussion	6